INCH-POUND

MIL-C-28754/81A
23 March 1994
SUPERSEDING
MIL-C-28754/81(NAVY)
21 April 1989

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, TYPE III, FEMALE CONTACT AND POLARIZING BUSHING SET, LOW INSERTION FORCE (LIF), REMOVABLE WIRE WRAPPOST TERMINAL

* This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-28754.

AMSC N/A
FSC 5935
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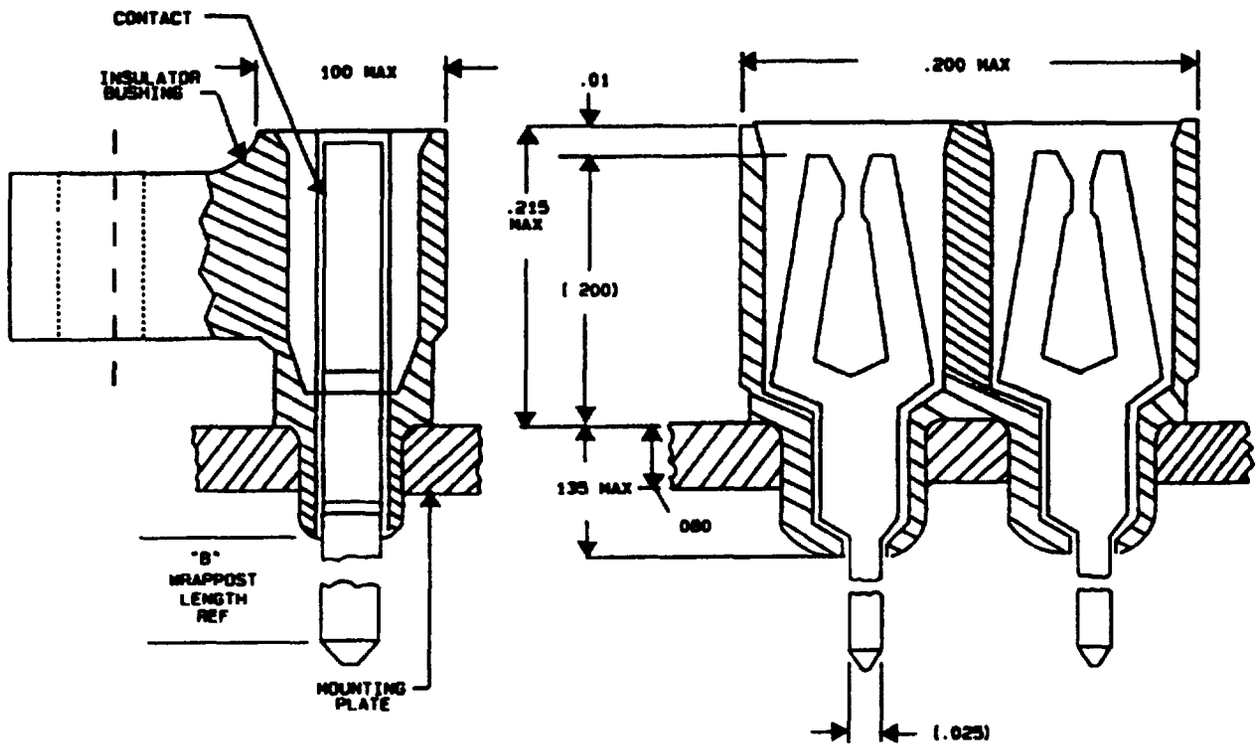


FIGURE 1. Typical female contact, polarizing bushing and mounting plate assembly.

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Inches	MM	Inches	MM
.010	.25	.135	3.43
.025	.64	.200	5.08
.080	2.03	.215	5.46
.100	2.54		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are plus or minus .005 inch (.13 mm) on three place decimals, plus or minus .01 inch (.3 mm) on two place decimals and plus or minus 2° on angles.

FIGURE 1. Typical female contact, polarizing bushing and mounting plate assembly - Continued.

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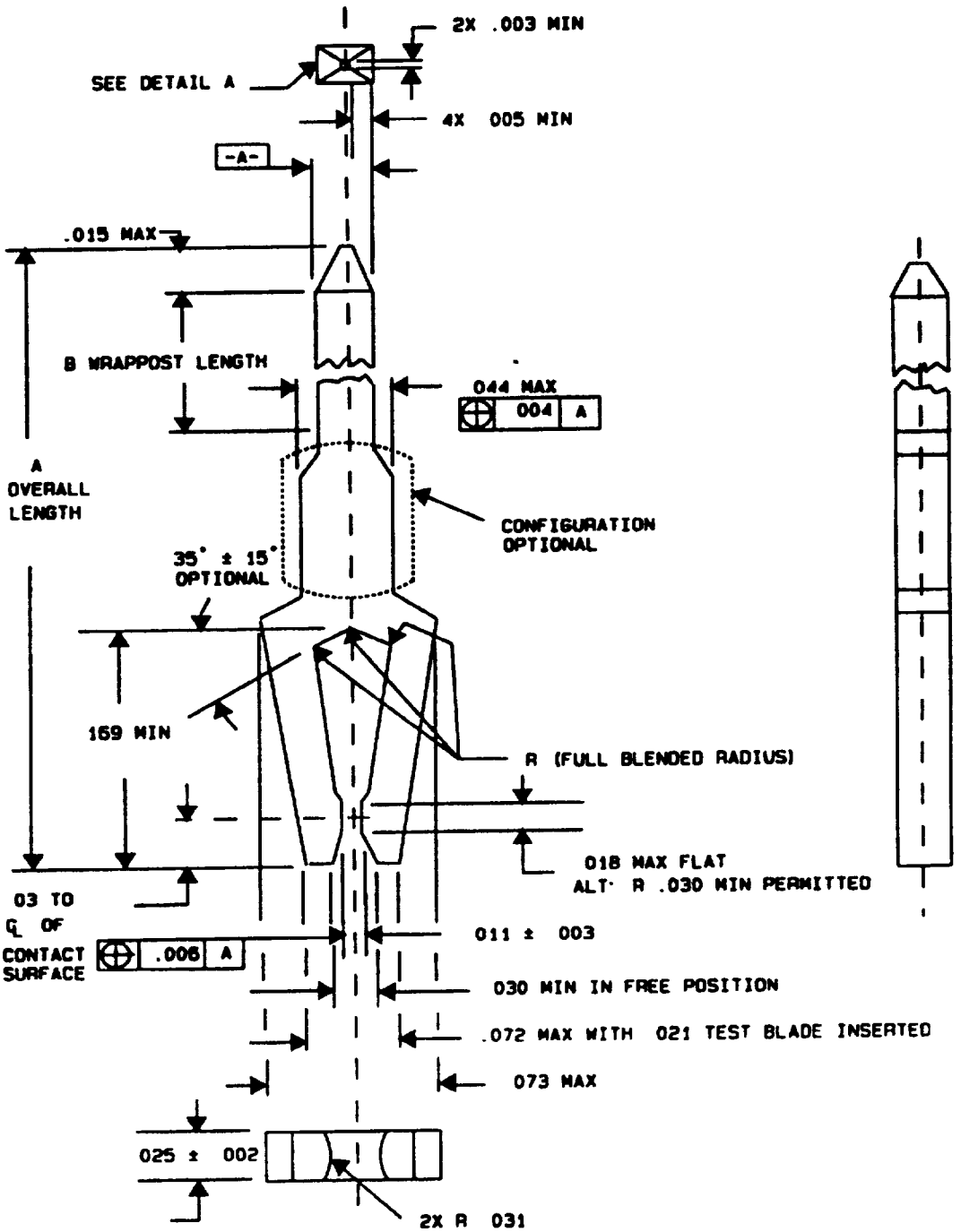
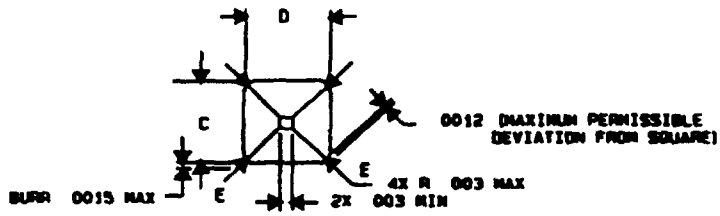


FIGURE 2. Contact, female, wrappost.

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DETAIL A

Wrappost Geometry				
C	D	E	Parallelism	Straightness
.025 (Nom) .022 (Min)	.025 (Nom) .022 (Min)	.0355 (Max) .0325 (Min)	.002	.005 in/in

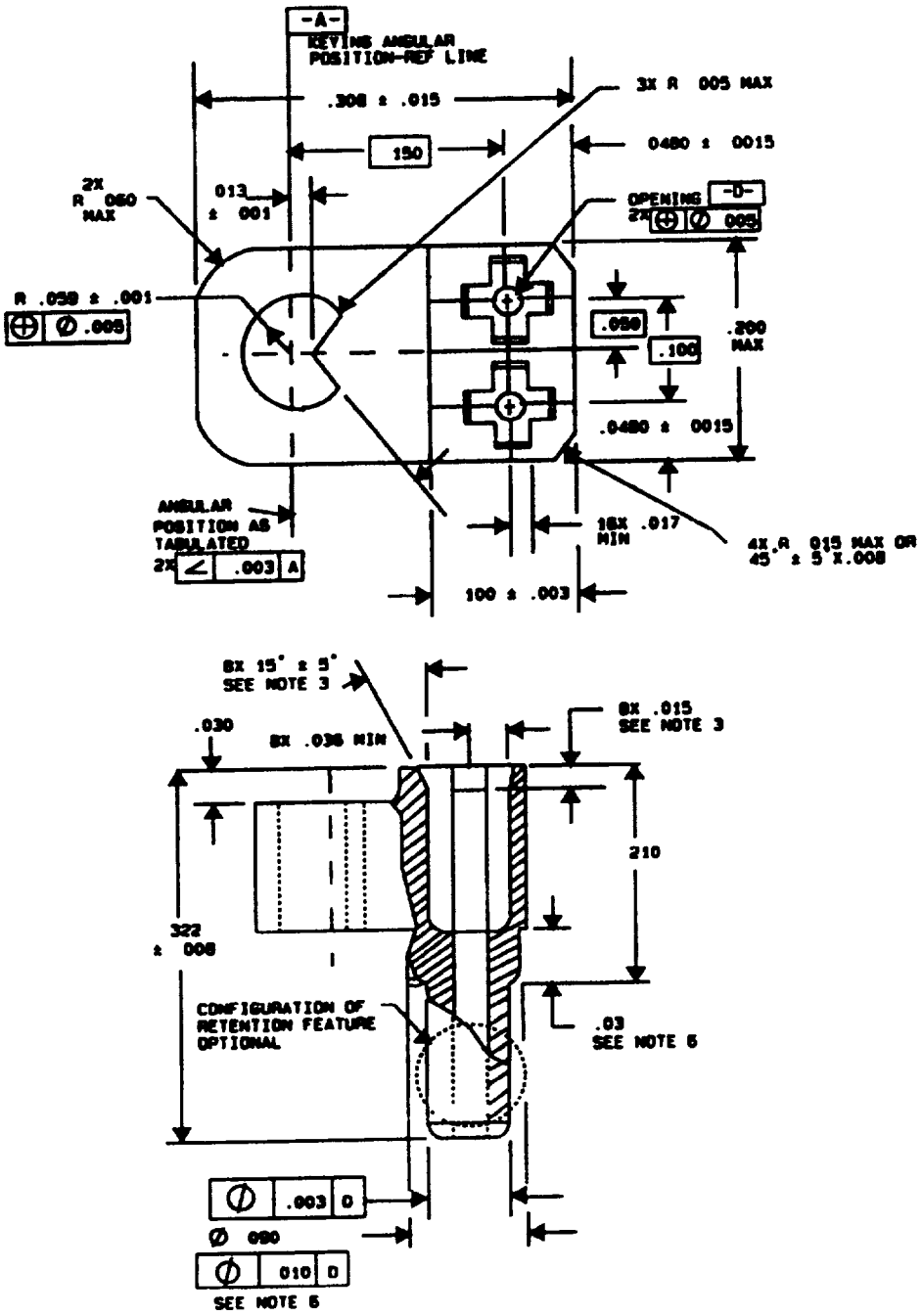
Inches	MM	Inches	MM	Inches	MM	Inches	MM
.002	.05	.010	.25	.030	.76	.164	4.17
.003	.08	.015	.38	.031	.79		
.004	.10	.018	.46	.044	1.12		
.005	.13	.021	.53	.072	1.83		
.006	.15	.025	.64	.073	1.85		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. See table I for dimensions A and B.
4. Unless otherwise specified, tolerances are plus or minus .005 inch (.13 mm) on three place decimals, plus or minus .01 inch (.3 mm) on two place decimals and plus or minus 2° on angles.

FIGURE 2. Contact, female, wrappost - Continued.

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OUTLINE A

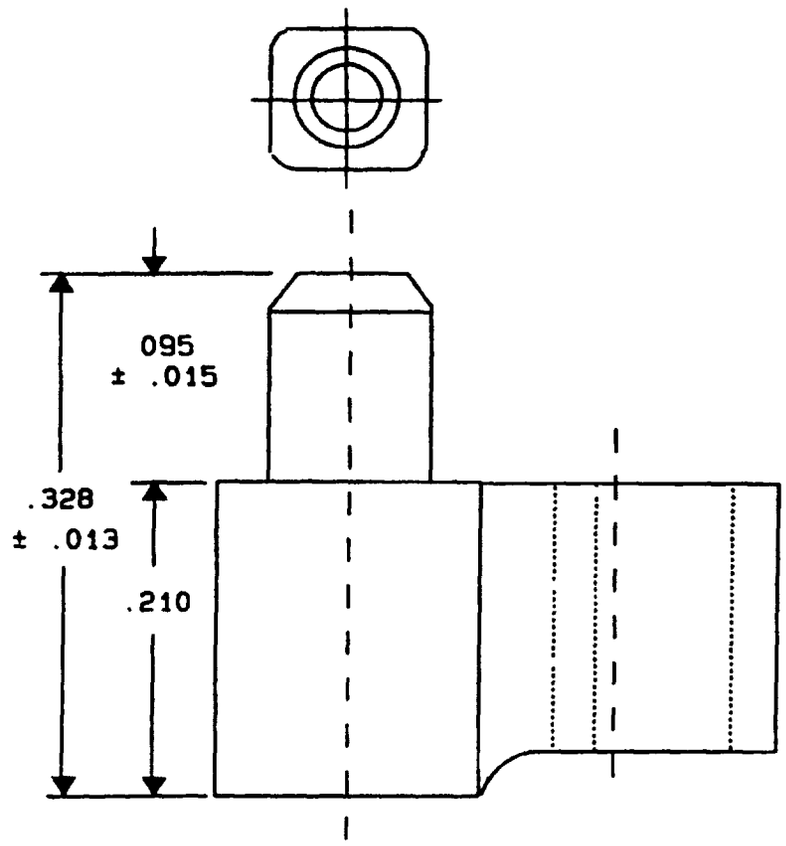
FIGURE 3(a). Bushing, polarizing, female.

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Inches	MM	Inches	MM	Inches	MM	Inches	MM
.001	.03	.015	.38	.060	1.52	.322	8.18
.0015	.038	.017	.43	.090	2.29		
.003	.08	.030	.76	.100	2.54		
.005	.13	.036	.91	.150	3.81		
.008	.20	.0480	1.219	.200	5.08		
.010	.25	.050	1.27	.210	5.33		
.013	.33	.059	1.50	.308	7.82		

FIGURE 3(a). Bushing, polarizing, female - Continued.

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ALTERNATE DESIGN

Inches	MM
.013	.330
.015	.381
.095	2.41
.210	5.334
.328	8.33

FIGURE 3(a). Bushing, polarizing, female - Continued.

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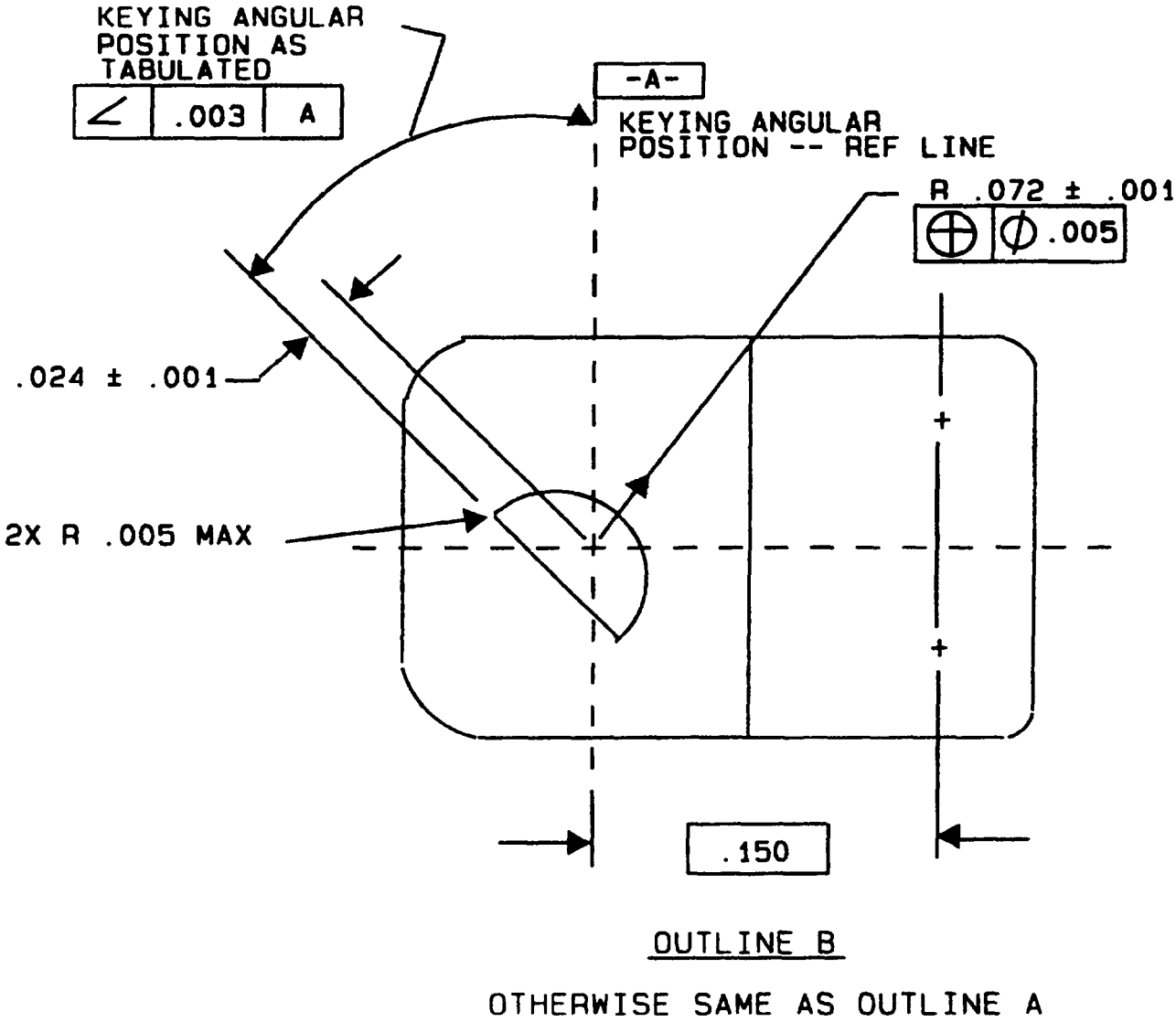


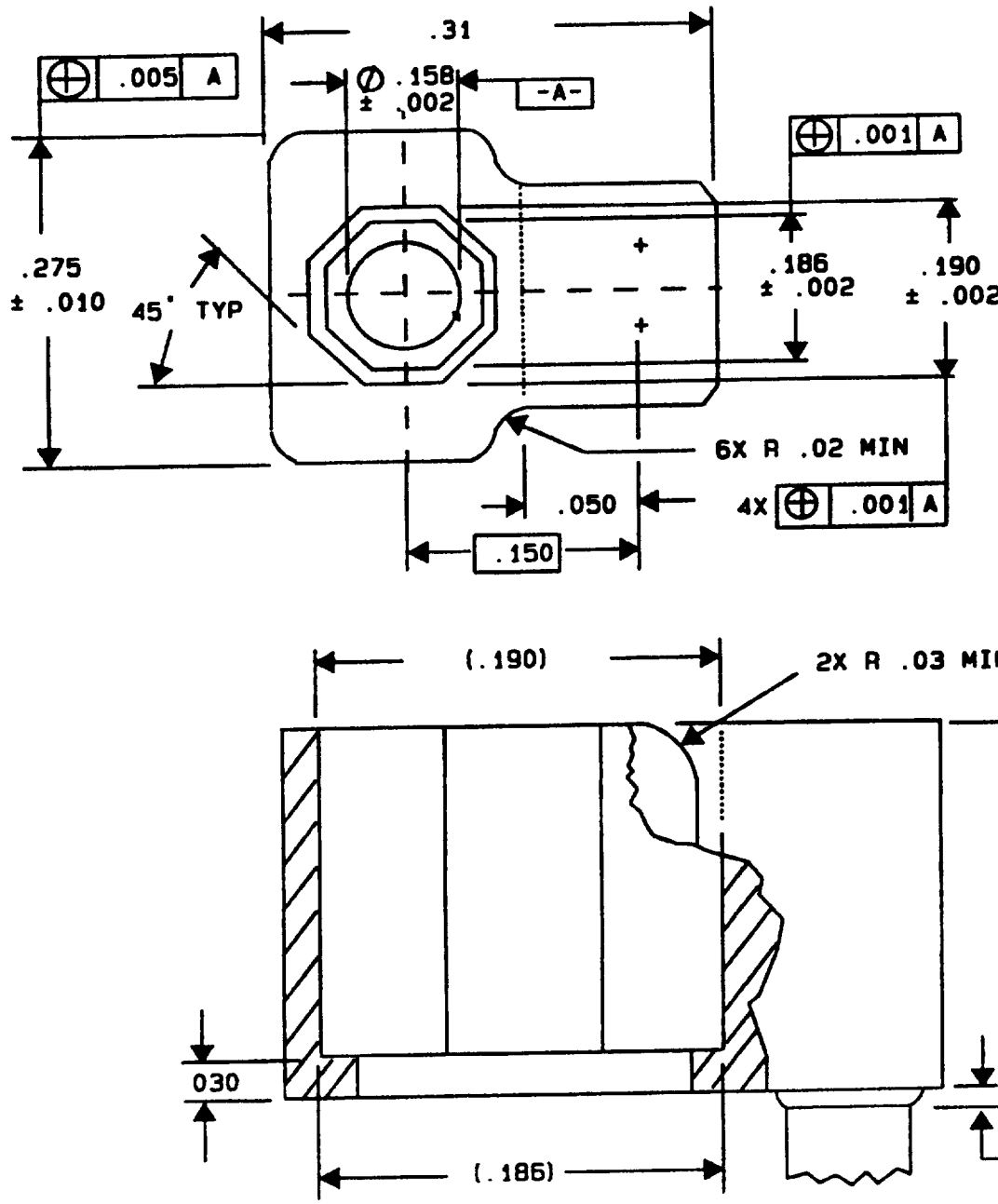
FIGURE 3(b). Bushing, polarizing, female.

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Inches	MM
.001	.03
.003	.08
.005	.13
.024	.61
.072	1.83
.150	3.81

FIGURE 3(b). Bushing, polarizing, female - Continued.

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OUTLINE C

OTHERWISE SAME AS OUTLINE A

FIGURE 3(c). Bushing, polarizing, female.

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Inches	MM	Inches	MM	Inches	MM	Inches	MM
.001	.03	.020	.51	.158	4.01	.275	6.98
.002	.05	.030	.76	.186	4.72	.31	7.9
.005	.13	.050	1.27	.190	4.83		
.010	.25	.150	3.81	.210	5.33		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Radius optional on .015 plus or minus .005 X 15° plus or minus 5° chamfer.
4. Dimensions include draft.
5. Unless otherwise specified, inside corners R .005 maximum.
6. The .03 X .090 feature is optional.
7. Unless otherwise specified, tolerances are plus or minus .005 inch (.13 mm) for three place decimals, plus or minus .01 inch (.3 mm) for two place decimals and plus or minus 2° on angles.

FIGURE 3(c). Bushing, polarizing, female - Continued.

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Polarizing style	Outline letter	Keying position as viewed at bushing's top surface
a	A	0° 
b	A	45° 
c	A	90° 
d	A	135° 
e	A	180° 
f	A	225° 
g	A	270° 
h	A	315° 
1	B	0° 

FIGURE 3(d). **Bushing, polarizing, female (keying hole orientation).**

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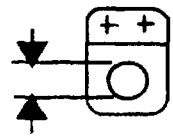
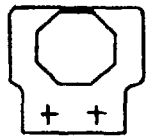
Polarizing style	Outline letter	Keying position as viewed at bushing's top surface
j	B	45° 
k	B	90° 
l	B	135° 
m	B	180° 
n	B	225° 
p	B	270° 
q	B	315° 
r	Same as 'A' except for thru hole as shown	Ø .093 ± 003 
s	C	

FIGURE 3(d). Bushing, polarizing, female (keying hole orientation) - Continued.

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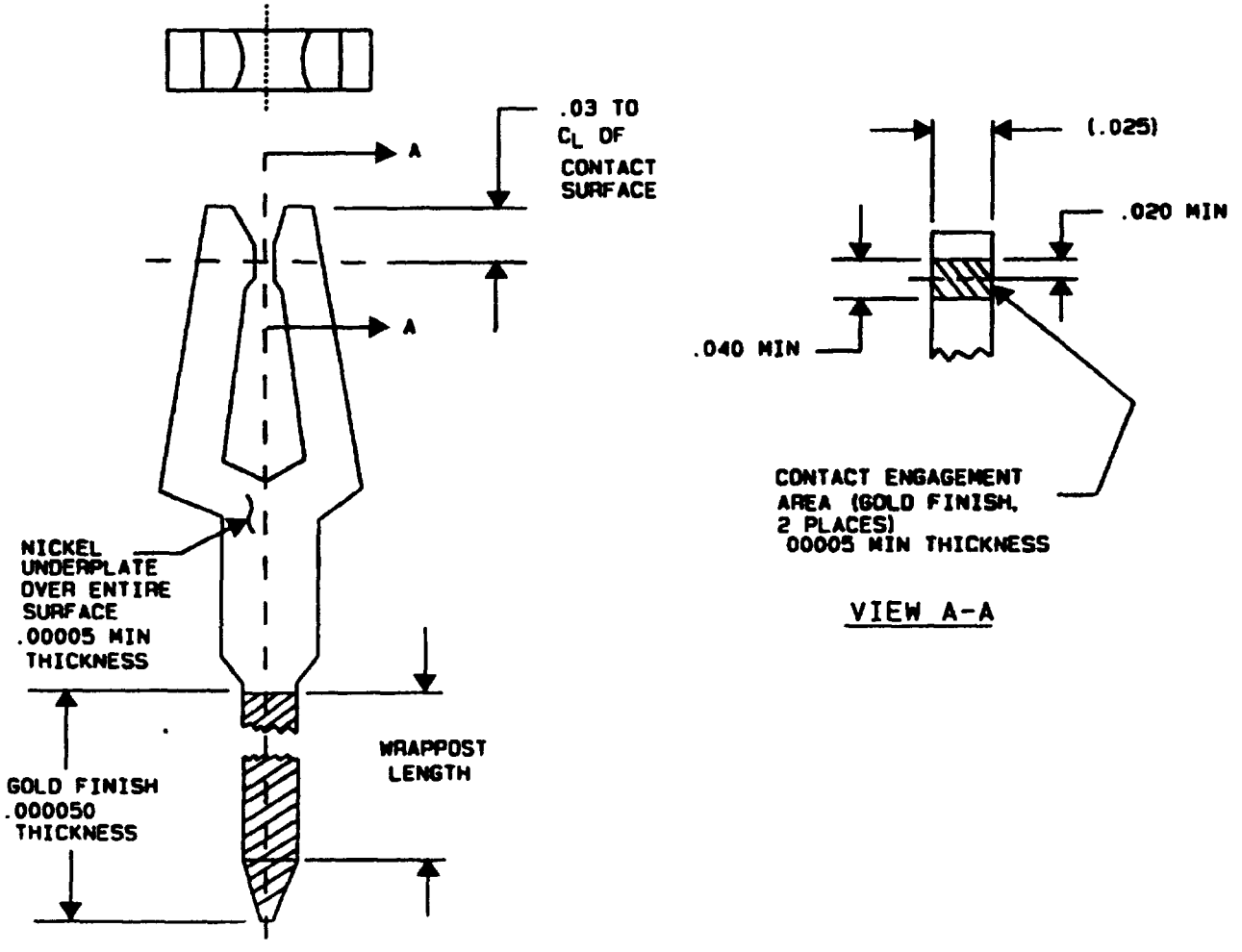


FIGURE 4. Localized finish (optional).

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Inches	MM	Inches	MM
.00005	.0013	.020	.51
.0001	.003	.025	.64
.0003	.008	.03	.8
.010	.25	.040	1.02

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are plus or minus .005 inch (.13 mm) on three place decimals, plus or minus .01 inch (.3 mm) on two place decimals and plus or minus 2° on angles.
4. Dimensions on this figure are for locating areas of localized finish only.

FIGURE 4. Localized finish (optional) - Continued.

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REQUIREMENTS:

Dimensions and configurations: See table I and figures 1, 2, 3 and 4.

Material:

*Contacts: Contacts shall be beryllium copper in accordance with ASTM B194, alloy UNS No. C17200, minimum hardness of 310HK.

Exposed base material: Exposed base material is permitted in noncritical breakoff or plating carrier areas of the connector parts providing all performance requirements are maintained with no degradation of critical areas.

*Bushing: Bushing shall be polyamide (nylon) per L-P-410 or ASTM D4066, group I (see figure 3).

Polarizing bushing color: The polarizing bushing color shall be the natural color of the base material.

Standard contact plating: Standard contact plating shall be in accordance with MIL-C-28754.

*Localized contact finish: Contacts may be supplied with a localized finish. Contacts supplied with a localized finish shall meet the requirements of MIL-C-28754 and this slash sheet. Contacts supplied with a localized finish shall conform to figures 2 and 4.

Gold finish: Gold shall be in accordance with MIL-G-45204, type II, grade C, class 1, and to a thickness of 50 micro-inches minimum. Area of gold finish shall be as specified on figure 4.

Nickel finish: Nickel finish shall be in accordance with QQ-N-290, class 1, and to a thickness of 50 microinches minimum over the entire contact surface (see figure 4).

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Performance:

Low level (dry circuit): Voltage drop shall not exceed 20 millivolts. Wire size shall be 26 AWG.

Contact resistance: Voltage drop shall not exceed 20 millivolts after durability cycling.

Current rating: 3 amperes.

Voltage rating: 300 volts, ac (rms) at sea level.

Temperature rating: -55°C to +125°C.

Marking: In accordance with MIL-C-28754.

Shipping: Bulk shipment shall be in unassembled condition.

Military part number M28754/81- (dash number from table I).

TABLE I. Dash numbers and dimensions.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
01	a	.513	.175
02	b	.513	.175
03	c	.513	.175
04	d	.513	.175
05	e	.513	.175
06	f	.513	.175
07	g	.513	.175
08	h	.513	.175
09	i	.513	.175

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TABLE I. Dash numbers and dimensions - Continued.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
10	j	.513	.175
11	k	.513	.175
12	l	.513	.175
13	m	.513	.175
14	n	.513	.175
15	p	.513	.175
16	q	.513	.175
17	r	.513	.175
18 1/	s	.513	.175
19	a	.588	.250
20	b	.588	.250
21	c	.588	.250
22	d	.588	.250
23	e	.588	.250
24	f	.588	.250
25	g	.588	.250
26	h	.588	.250
27	i	.588	.250
28	j	.588	.250
29	k	.588	.250

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TABLE I. Dash numbers and dimensions - Continued.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
30	l	.588	.250
31	m	.588	.250
32	n	.588	.250
33	p	.588	.250
34	q	.588	.250
35	r	.588	.250
36 1/	s	.588	.250
37	a	.688	.350
38	b	.688	.350
39	c	.688	.350
40	d	.688	.350
41	e	.688	.350
42	f	.688	.350
43	g	.688	.350
44	h	.688	.350
45	i	.688	.350
46	j	.688	.350
47	k	.688	.350
48	l	.688	.350
49	m	.688	.350
50	n	.688	.350

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TABLE I. Dash numbers and dimensions - Continued.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
51	p	.688	.350
52	q	.688	.350
53	r	.688	.350
54 1/	s	.688	.350
55	a	.838	.500
56	b	.838	.500
57	c	.838	.500
58	d	.838	.500
59	e	.838	.500
60	f	.838	.500
61	g	.838	.500
62	h	.838	.500
63	i	.838	.500
64	j	.838	.500
65	k	.838	.500
66	l	.838	.500
67	m	.838	.500
68	n	.838	.500
69	p	.838	.500
70	q	.838	.500
71	r	.838	.500
72 1/	s	.838	.500

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TABLE I. Dash numbers and dimensions - Continued.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
73	a	.963	.625
74	b	.963	.625
75	c	.963	.625
76	d	.963	.625
77	e	.963	.625
78	f	.963	.625
79	g	.963	.625
80	h	.963	.625
81	i	.963	.625
82	j	.963	.625
83	k	.963	.625
84	l	.963	.625
85	m	.963	.625
86	n	.963	.625
87	p	.963	.625
88	q	.963	.625
89	r	.963	.625
90 1/	s	.963	.625
91	a	1.088	.750
92	b	1.088	.750
93	c	1.088	.750

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TABLE I. Dash numbers and dimensions - Continued.

Dash No	Polarizing style (fig 3d)	Overall length "A" ± 0.010	Wrappost length "B" (Ref)
94	d	1.088	.750
95	e	1.088	.750
96	f	1.088	.750
97	g	1.088	.750
98	h	1.088	.750
99	i	1.088	.750
100	j	1.088	.750
101	k	1.088	.750
102	l	1.088	.750
103	m	1.088	.750
104	n	1.088	.750
105	p	1.088	.750
106	q	1.088	.750
107	r	1.088	.750
108 1/	s	1.088	.750

1/ Keying pegs for insertion into the octal hole are covered by MIL-C-28754/39.

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Supersession data: This specification sheet includes the requirements of drawing 5516568/21 through 22, code identification 53711.

QUALIFICATIONS: See appendix.

Localized contact finish: The qualification of the contact supplied with the standard finish does not extend qualification to the contact supplied with the localized finish. To be qualified to supply the localized finish in lieu of the standard contact finish, manufacturers shall subject the contact supplied with localized finish to the same qualification procedures as the standard contact finish. Manufacturers that have qualified the localized finish contact will be designated on the Qualified Products List (QPL).

*Changes from previous issue: The margins of this specification sheet are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - CR
Navy - NW
Air Force - 85

Review activities:
Navy - MC
Air Force - 11, 99
DLA - ES

User activities:
Army - AR, MI
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-28754/81A	1994 March 23

3. DOCUMENT TITLE: Connectors, Electrical, Modular, Type III, Female Contact and Polarizing Bushing Set, Low Insertion Force (LIF), Removable Wire Wrappost Terminal

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-5993	88-482-5993
c. ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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